LEPAn app

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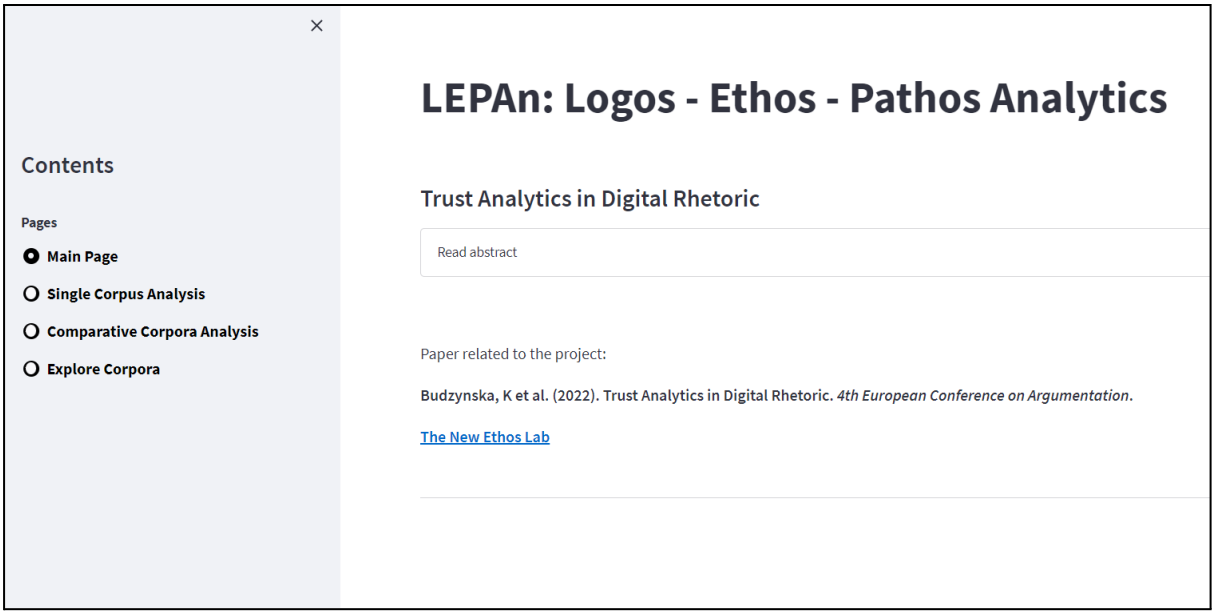
"Main Page"

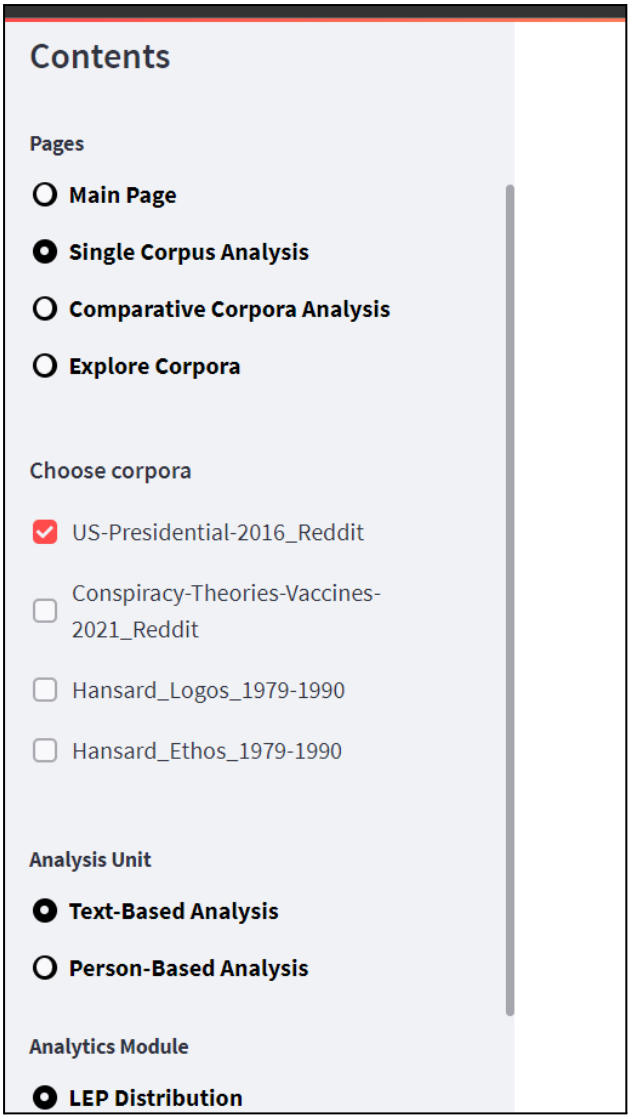


It shows basic information about the project. You can read the abstract of the *Trust* paper (by clicking on the "*Read abstract*").

On the left hand side, you can see a navigation menu – 4 pages are displayed. You go to the selected page by clicking on the name of the page.

"Single Corpus Analysis"

By clicking on the *"Single Corpus Analysis"*, you will see the following navigation menu on the left hand side:



By default you will be taken to the *"Text-Based Analysis"* Analysis Unit and the *"LEP Distribution"* module.

SUBPAGES

ANALYSIS UNITS

First, you need to choose the corpus by “ticking” the square next to the name of the corpus.
In the *"Single Corpus Analysis"* you have 4 option for corpora:

- *"US-Presidential-2016_Reddit"*,
- *"Conspiracy-Theories-Vaccines-2021_Reddit"*,
- *Hansard_Logos_1979-1990*,

- *Hansard_Ethos_1979-1990*.

By default the *"US-Presidential-2016_Reddit"* corpus is chosen.

You can choose 1, 2, or 3 corpora – regarding the *Hansard_1979-1990* corpus, you can choose either *Hansard_Logos_1979-1990* or *Hansard_Ethos_1979-1990* at the same time.

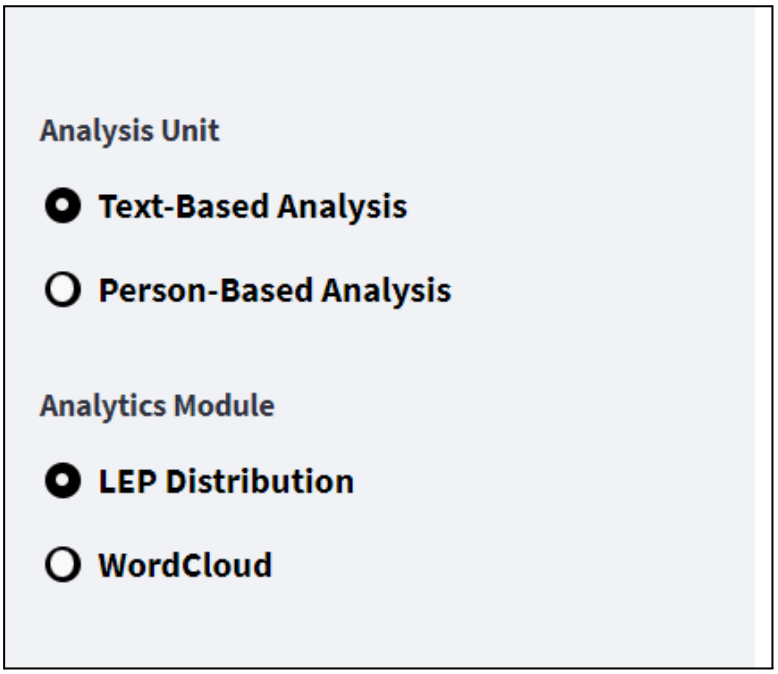
Second, you need to choose the *Analysis Unit*.
In the *"Single Corpus Analysis"* you can choose between two *Analysis Units*:

- *"Text-Based Analysis"*
- "*Person-Based Analysis"*.

"Text-Based Analysis"

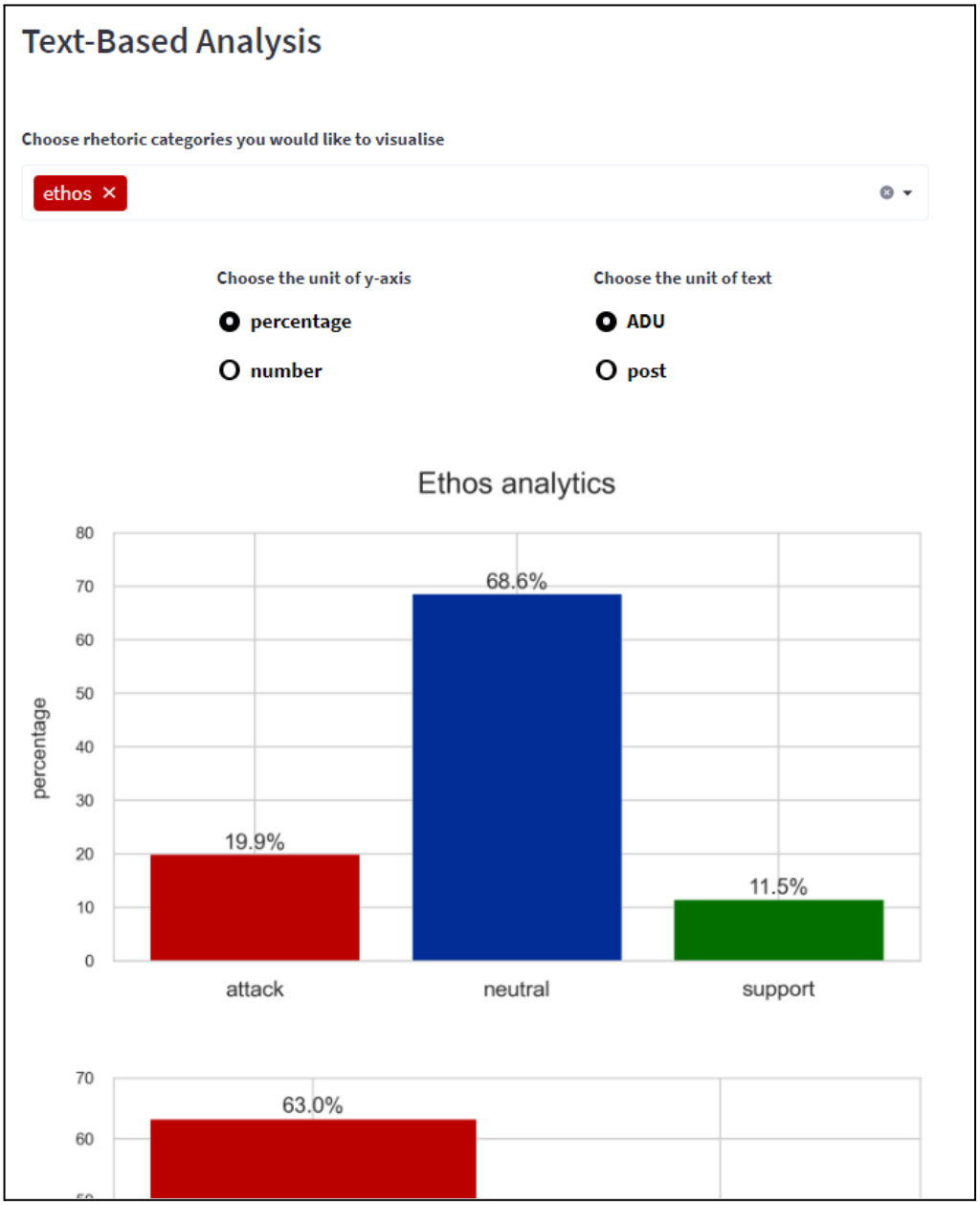
In this *Analysis Unit* you can choose between the following 2 modules:

- *"LEP Distribution"*
- *"WordCloud"*.



ANALYTICS MODULES

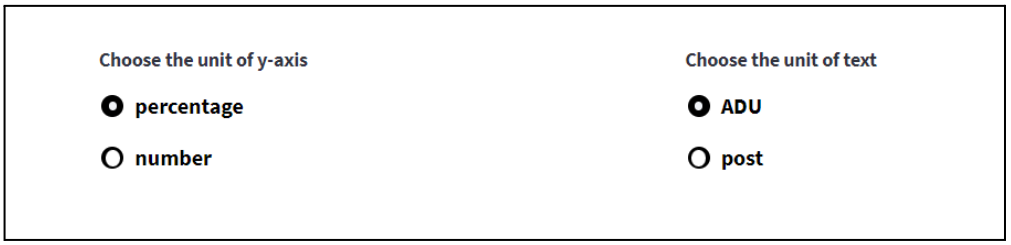
"LEP Distribution"



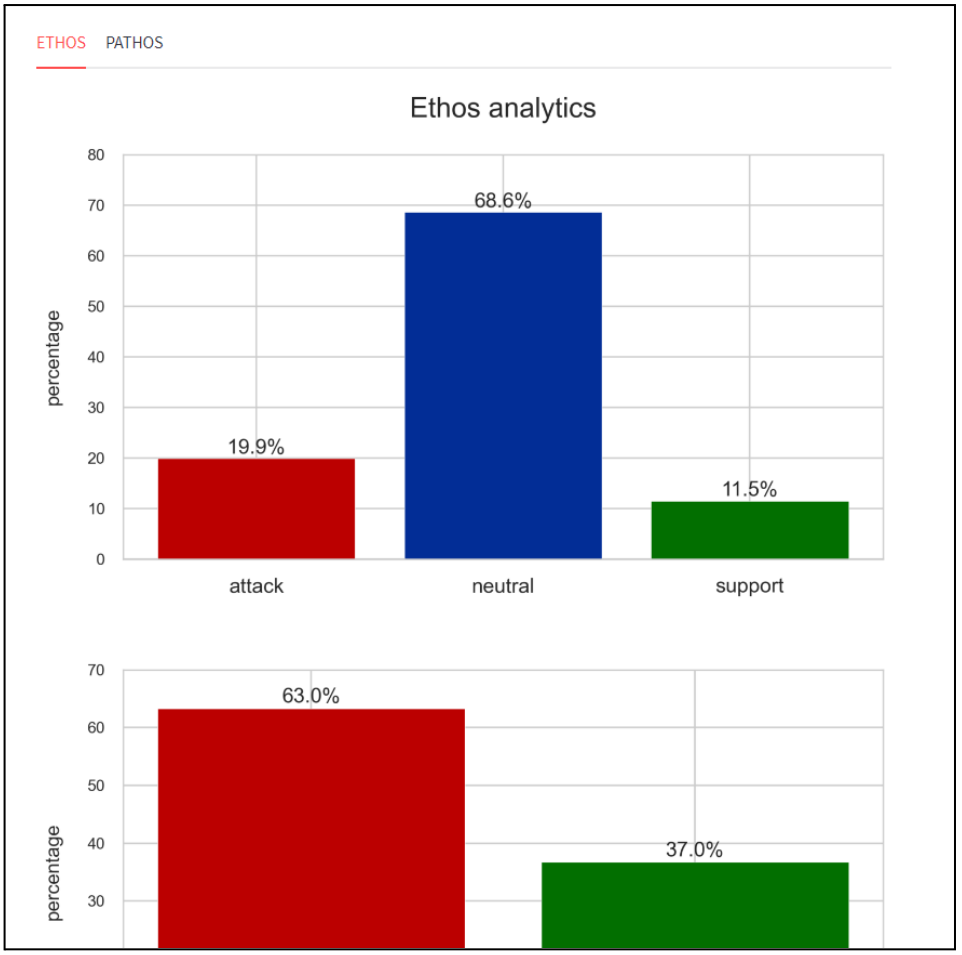
It shows the distribution of the chosen rhetoric category(-ies): logos, ethos and/or pathos in the selected corpora.

“Choose rhetoric categories you would like to visualise” — by clicking on the element (container) below this caption you will see a list with categories of rhetoric you can choose from. You can choose 1, 2 or all 3 options. In red rectangles you will see the categories you have chosen. By default the *“ethos”* category is chosen. Later on you will see the distribution of subcategories of the chosen rhetoric category.

After choosing rhetoric categories, you can decide the unit of y-axis ("**Choose the unit of y-axis**") and the unit of text ("**Choose the unit of text**") for the analysis and visualisations. Depending on the chosen corpora, options available for the unit of text can change. **ADU** stands for *Argumentative Discourse Unit*. **Post** is the original post from social media (Twitter or Reddit) or full utterance (in the case of *Hansard* corpora).



Finally, you can see the distribution of subcategories (attack/negative, support/positive, neutral) for the chosen rhetoric category(-ies), and text and y-axis units. In addition in the upper left corner in the picture below you can see the tabs with the names of rhetoric categories you have chosen. You can switch between the tabs (by clicking on the name of a category in upper case) to see visualisations for each category.


ETHOS
PATHOS
Ethos analytics
percentage
80
70
60
50
40
30
20
10
0
19.9%
68.6%
11.5%
attack
neutral
support
percentage
70
60
50
40
30
63.0%
37.0%

"WordCloud"

Visualise words with high precision for rhetoric categories.

First, you choose a rhetoric category (logos, ethos or pathos), and a subcategory (attack/negative, support/positive, both). By default the logos category and attack/negative subcategory are chosen.

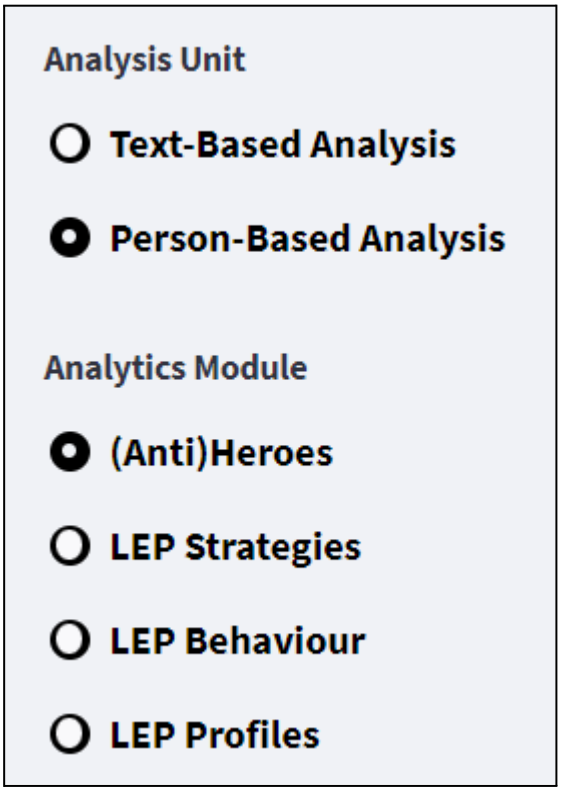
Second, choose a precision value for word-cloud. By default 90 is chosen as the precision value. The higher the value, the more “predictive” the words you will see for the chosen category and subcategory in the selected corpora.

"Person-Based Analysis"

In this subpage you can choose 1 out of the following 4 modules:

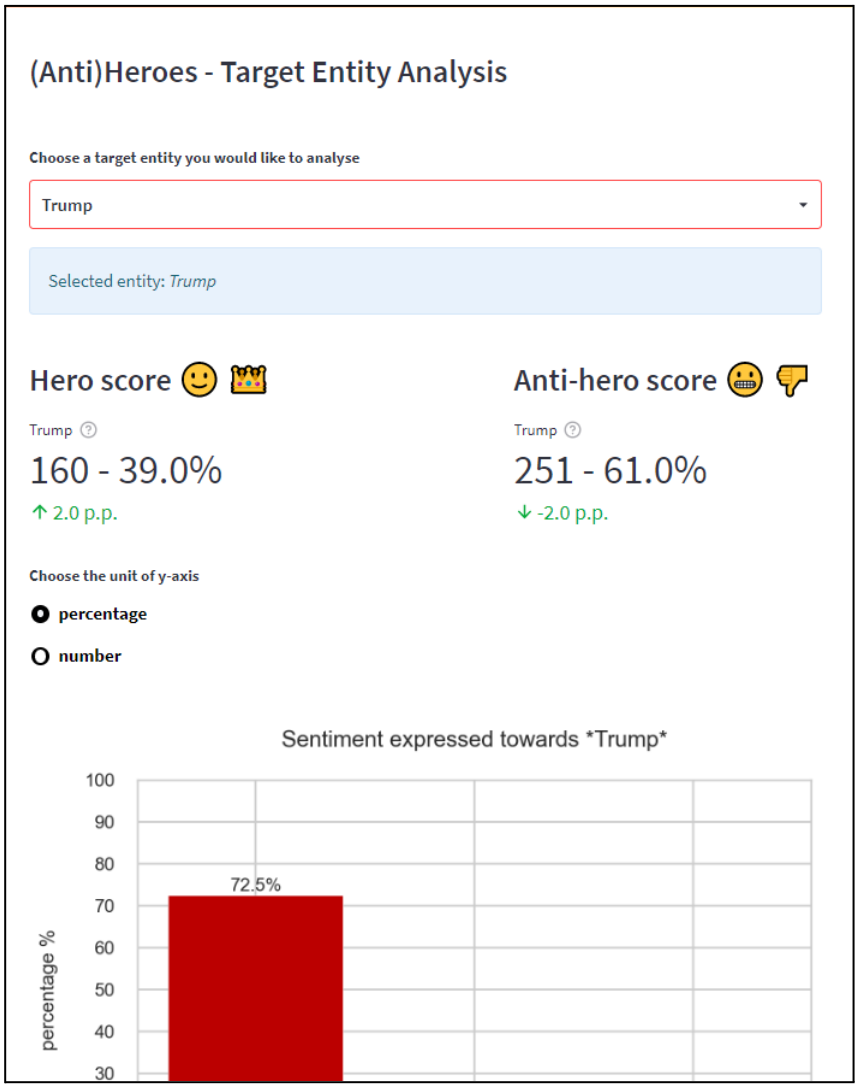
- *"(Anti)Heroes"*,

- *"LEP Strategies"*,
- *"LEP Behaviour"*,
- *"LEP Profiles"*.



ANALYTICS MODULES

"(Anti)Heroes"



It shows the (anti)hero metrics for the chosen target entity. Metrics based on the ethos annotation – what percentage of people view him or her as hero and anti-hero (that is, support or attack this person in terms of ethos).

First, you need to choose the name of the target entity from the list that is expanded after clicking on it ("**Choose a target entity you would like to analyse**").

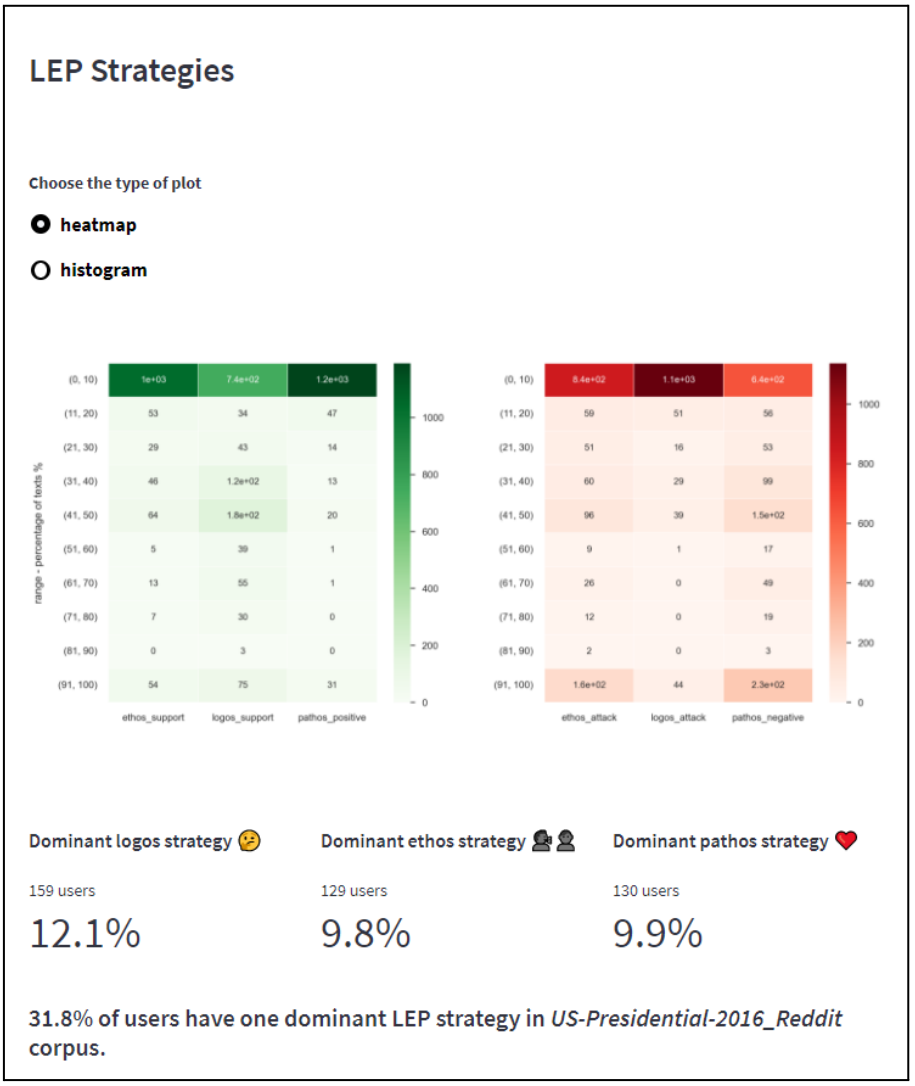
You will see the *Hero score* (on the left hand side of the page) and *Anti-hero score* (on the right hand side of the page) for the chosen entity. In both scores, the first number (160 and 251 in the example) indicates the number of texts that support or attack, respectively, a chosen entity in terms of ethos, and the second number (39 and 61 in the example) indicate the

percentage of texts in the whole corpora that support or attack, respectively, a chosen entity in terms of ethos.

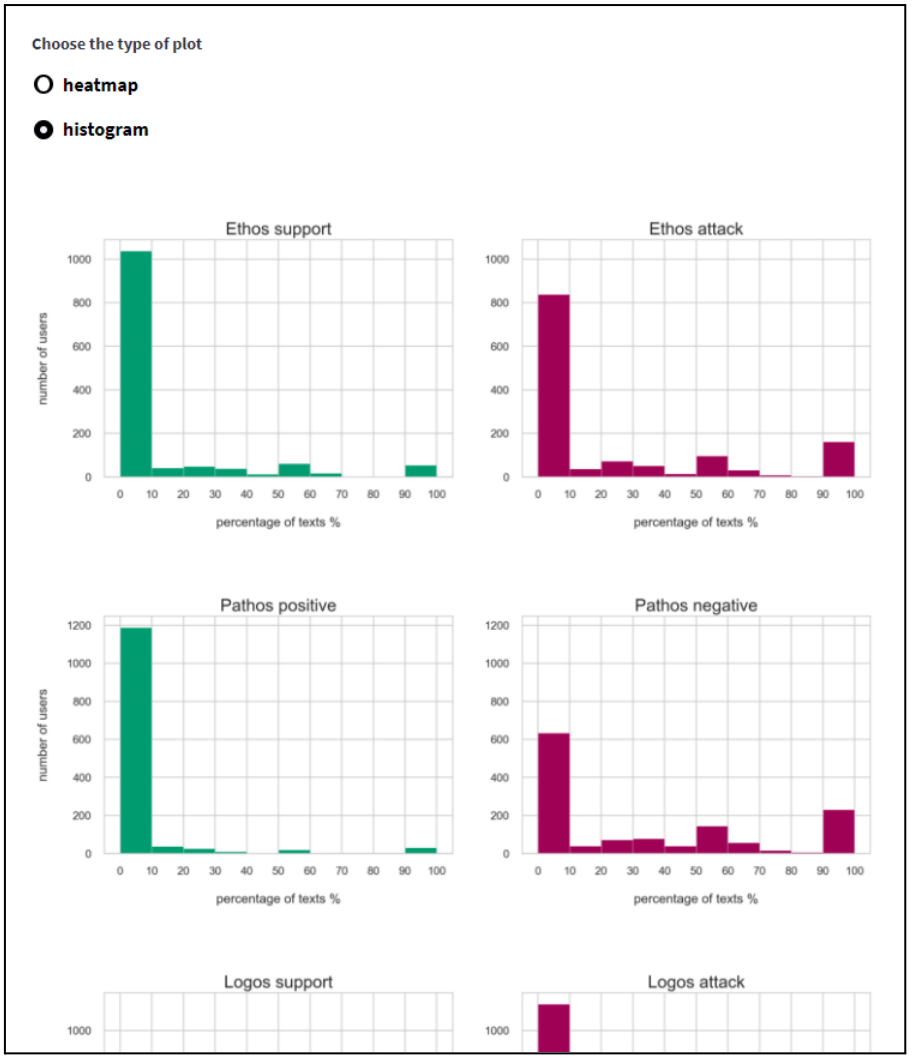
Secod, you can choose the unit of y-axis ("**Choose the unit of y-axis**") for visualisation of the distribution of sentiment expressed towards the chosen entity.

Finally, you will see the distribution.

"LEP Strategies"



Heatmap.



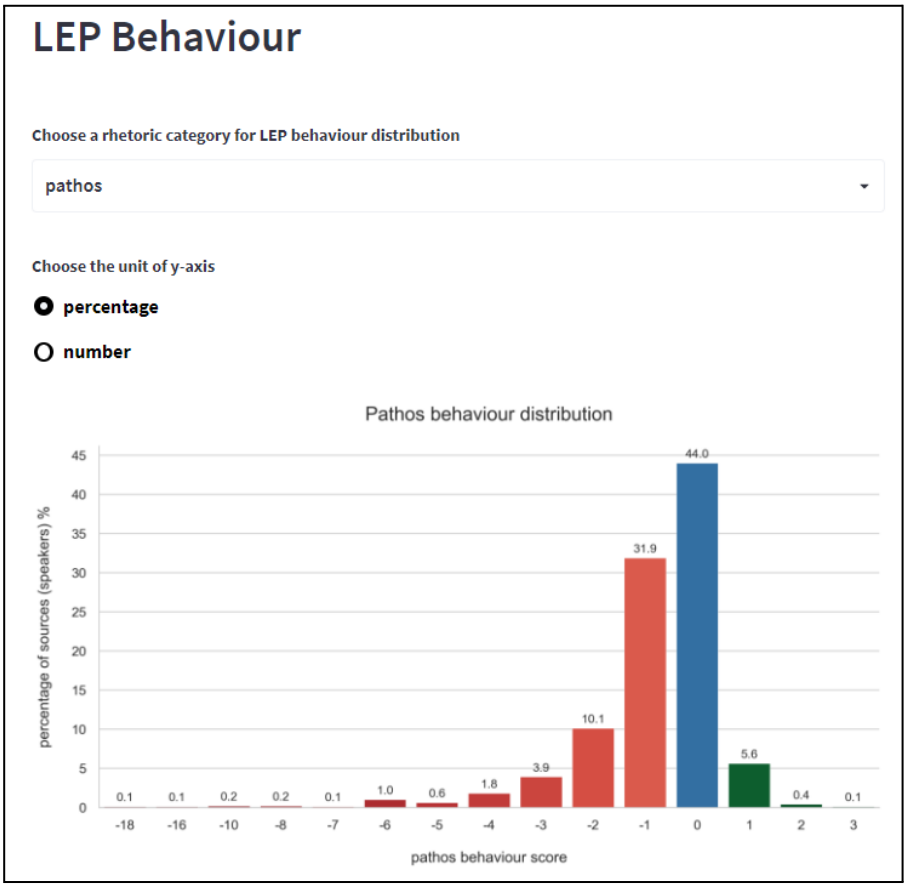
Histogram.

LEP Strategy – for each user *u* it calculates the percentage of texts generated by user *u* that are classified as logos attacks, logos supports, pathos negative, etc. Then, it shows the distribution of those calculated proportions in the form of a histogram or a heatmap.

In the example below (in the image attached) in terms of ethos support (strategy) for most users this statistic is between 0 and 10%. That is, for most users ethotic supports comprise between 0 and 10% of their respective texts.

Dominant strategies – it is the percentage of users for which the proportion of texts classified as logos, ethos, and pathos is above the mean plus one standard deviation (i.e., percentage of users who employ logos, ethos or pathos significantly above the average).

"LEP Behaviour"



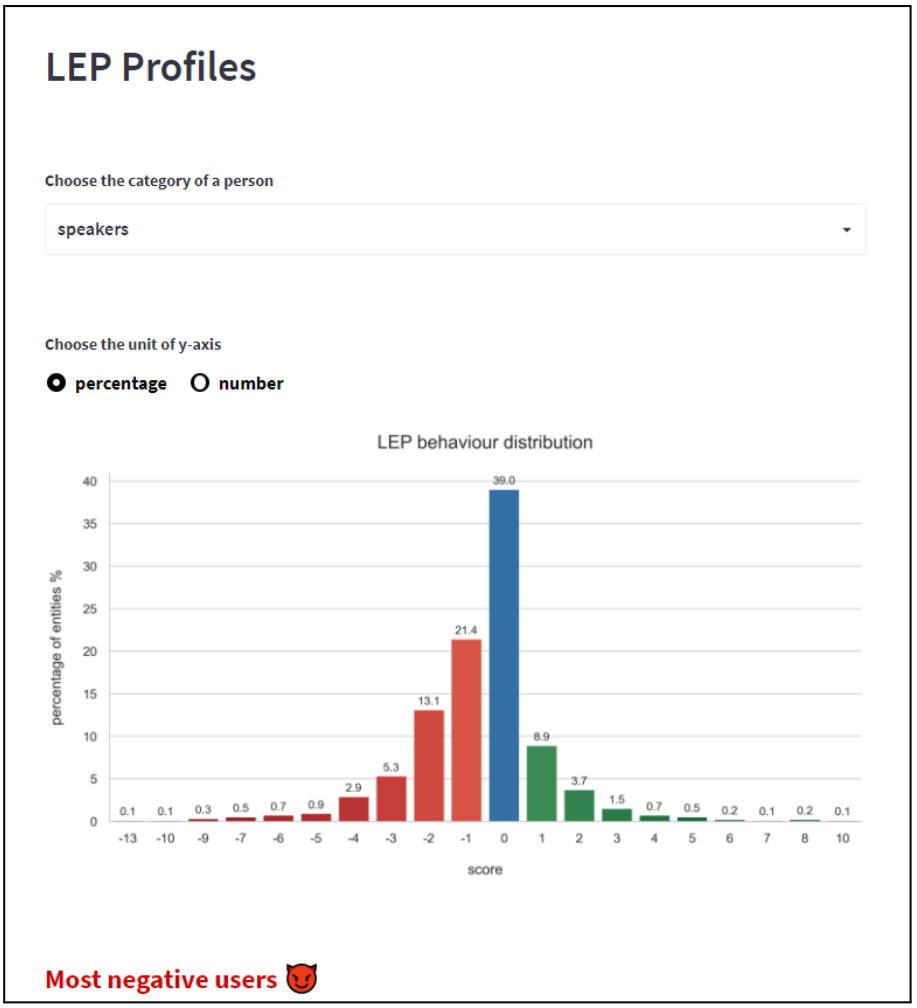
It shows the distribution of the LEP behaviour metric for the chosen category of logos, ethos or pathos.

LEP behaviour metric is based on a simple algorithm for calculating the “negativity” (values below zero; red bars in a plot), “positivity” (values above zero; green bars in a plot) or “neutrality” (values equal to zero; blue bars in a plot) for each user based on the number of texts generated by each user and classified as logos attacks and logos supports, or pathos negative and pathos positive, or ethos attacks and ethos supports.

"LEP Profiles"

First, it shows the distribution of the aggregated LEP behaviour metric. Similar to the distribution displayed in the **"LEP Behaviour"** module, however now all rhetoric categories are

considered for calculating the final score (not separately for logos, ethos and pathos).



Then, based on this aggregated LEP behaviour metric it displays the most negative and positive users / entities together with the users' scores on this metric.

Most negative users 😈

TRUMP	No_stop_signs	MR_TELEVOID	Sryzon
LEP behaviour score	LEP behaviour score	LEP behaviour score	LEP behaviour score
-13	-10	-9	-9
↓ -6.8 SD	↓ -5.1 SD	↓ -4.6 SD	↓ -4.6 SD

stagshore	oldmanbrownsocks	crispy111	WineAndWhiskey
LEP behaviour score	LEP behaviour score	LEP behaviour score	LEP behaviour score
-9	-9	-7	-7
↓ -4.6 SD	↓ -4.6 SD	↓ -3.5 SD	↓ -3.5 SD

Choose name to see details about the user

● **TRUMP** ○ **No_stop_signs** ○ **MR_TELEVOID** ○ **Sryzon** ○ **stagshore**

○ **oldmanbrownsocks** ○ **crispy111** ○ **WineAndWhiskey**

Texts posted by: **TRUMP**

	Text	Target	pathos_name	ethos_name	logos_na
TRUMP	Only Rosie O'Donnell	<NA>	neutral	neutral	neutral
TRUMP	Megyn, you ignorant slut	<NA>	negative	neutral	neutral
TRUMP	I'd fuck my daughter over you	<NA>	negative	neutral	neutral
TRUMP	BAM I'll build a wall based on the statement of some guy	<NA>	negative	neutral	neutral
TRUMP	BAM Women's issues aren't issues	<NA>	negative	neutral	neutral
TRUMP	You have no idea how many people I've bribed	<NA>	negative	neutral	neutral

Lastly, it shows examples of text generated by a chosen negative or positive user and statistics for the user's use of logos, ethos and pathos strategy.

Choose name to see details about the user

- (●) TRUMP
- ( ) No_stop_signs
- ( ) MR_TELEVOID
- ( ) Sryzon
- ( ) stagshore
- ( ) oldmanbrownsocks
- ( ) crispy111
- ( ) WineAndWhiskey

Texts posted by: TRUMP

	Text	Target	pathos_name	ethos_name	logos_na
TRUMP	Only Rosie O'Donnell	<NA>	neutral	neutral	neutral
TRUMP	Megyn, you ignorant slut	<NA>	negative	neutral	neutral
TRUMP	I'd fuck my daughter over you	<NA>	negative	neutral	neutral
TRUMP	BAM I'll build a wall based on the statement of some guy	<NA>	negative	neutral	neutral
TRUMP	BAM Women's issues aren't issues	<NA>	negative	neutral	neutral
TRUMP	You have no idea how many people I've bribed	<NA>	negative	neutral	neutral
TRUMP	Why are you poor	<NA>	negative	neutral	neutral
TRUMP	I robbed three banks on my way to this public broadcast	<NA>	negative	neutral	support
TRUMP	no one would be bringing it up if he didn't say anything	<NA>	neutral	neutral	neutral
TRUMP	SIT DOWN AND SHUT UP	<NA>	negative	neutral	neutral

TRUMP's LEP strategy

Logos strategy	Ethos strategy	Pathos strategy
TRUMP	TRUMP	TRUMP
5.5	0.0	34.5
↓ -0.4 p.p.	↓ -20.0 p.p.	↑ 4.1 p.p.

TRUMP's negativity comes mostly from pathos

Most positive users 😀

ChristianTaz	MutthaFuzza	DeceitFive9	Cudder_Buddy
LEP behaviour score	LEP behaviour score	LEP behaviour score	LEP behaviour score
10	8	8	7
↑ 5.8 SD	↑ 4.7 SD	↑ 4.7 SD	↑ 4.2 SD

Helicase21	rowdyroddypiperjr	elkab0ng	Speedson
LEP behaviour score	LEP behaviour score	LEP behaviour score	LEP behaviour score
6	6	5	5
↑ 3.6 SD	↑ 3.6 SD	↑ 3.1 SD	↑ 3.1 SD

Choose name to see details about the user

- (●) **ChristianTaz**
- ( ) **MutthaFuzza**
- ( ) **DeceitFive9**
- ( ) **Cudder_Buddy**
- ( ) **Helicase21**
- ( ) **rowdyroddypiperjr**
- ( ) **elkab0ng**
- ( ) **Speedson**

Texts posted by: ChristianTaz

	Text	Target	pathos_name	ethc
ChristianTaz	Isn't that kind of the same as Bernie Sanders	Sanders	neutral	atta
ChristianTaz	As president, Sanders has promised to raise minimum wage to $15.00	<NA>	neutral	neut
ChristianTaz	make sure no one working 40 hours a week is in poverty	<NA>	neutral	neut
ChristianTaz	ut 13 million to work by investing a trillion dollars in infrastructure	<NA>	neutral	neut

Choose name to see details about the user

- (●) **ChristianTaz**
- ( ) **MutthaFuzza**
- ( ) **DeceitFive9**
- ( ) **Cudder_Buddy**
- ( ) **Helicase21**
- ( ) **rowdyroddypiperjr**
- ( ) **elkab0ng**
- ( ) **Speedson**

Texts posted by: **ChristianTaz**

	Text	Target	pathos_name	ethc
ChristianTaz	Isn't that kind of the same as Bernie Sanders	Sanders	neutral	atta
ChristianTaz	As president, Sanders has promised to raise minimum wage to $15.00	<NA>	neutral	neut
ChristianTaz	make sure no one working 40 hours a week is in poverty	<NA>	neutral	neut
ChristianTaz	ut 13 million to work by investing a trillion dollars in infrastructure	<NA>	neutral	neut
ChristianTaz	reverse the NAFTA, CAFTA and PNTR trade agreements	<NA>	neutral	neut
ChristianTaz	create a million jobs by investing 5.5 billion in youth jobs	<NA>	positive	neut
ChristianTaz	make tuition free at all public colleges	<NA>	neutral	neut
ChristianTaz	expand social security	<NA>	neutral	neut
ChristianTaz	enact universal healthcare and prekindergarten program	<NA>	neutral	neut
ChristianTaz	break up big finanacial institutions	<NA>	positive	neut

ChristianTaz's LEP strategy

Logos strategy	Ethos strategy	Pathos strategy
ChristianTaz	ChristianTaz	ChristianTaz
87.5	0.0	16.7
↑ 54.9 p.p.	↓ -11.6 p.p.	↑ 12.2 p.p.

ChristianTaz's positivity comes mostly from **logos**

"Comparative Corpora Analysis"

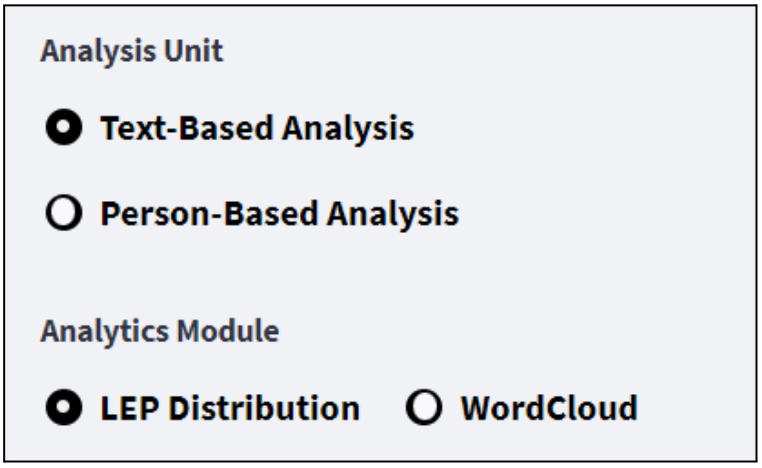
SUBPAGES

ANALYSIS UNITS

"Text-Based Analysis"

In this subpage you can choose between the following 2 modules:

- *"LEP Distribution"*
- *"WordCloud"*.



ANALYTICS MODULES

"LEP Distribution"

It shows distributions of logos, ethos and pathos in the selected set of corpora, next to each other for comparative purposes.

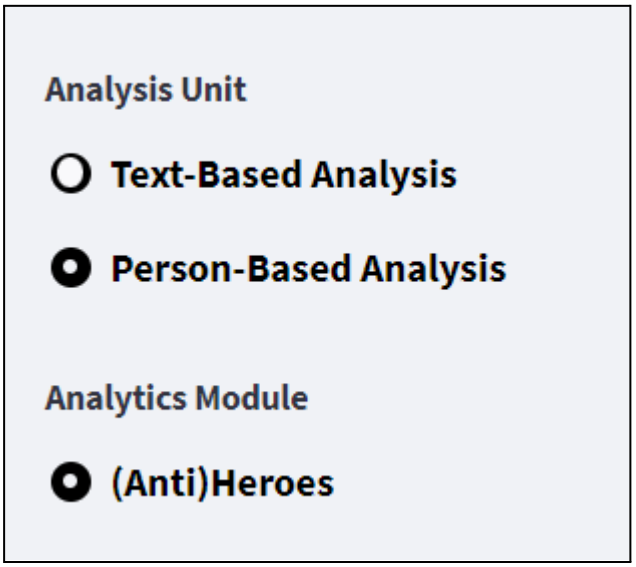
Thus, it is similar to single corpus analysis **"LEP Distribution"**.

"WordCloud"

It shows words equal or greater than a selected precision value for the selected subcategory of logos, ethos or pathos in all chosen corpora. It is thus similar for single corpus analysis **"WordCloud"**.

"Person-Based Analysis"

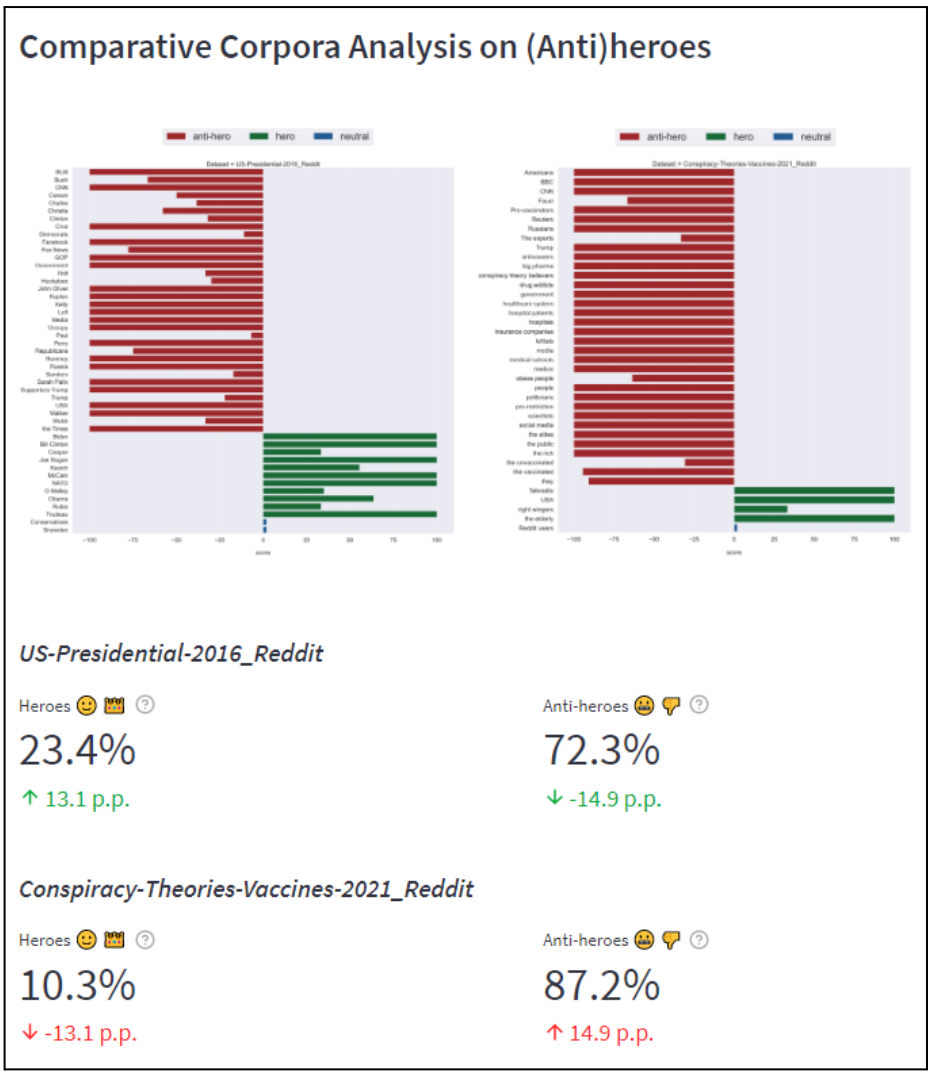
There is only one module – *"(Anti)Heroes"* – you will be directed to by default.



ANALYTICS MODULES

"(Anti)Heroes"

It shows (anti)heroes in the selected set of corpora. (Anti)Hero scores are calculated based on the ethos annotation – percentage of people that view a given entity as an anti-hero is subtracted from the percentage of people that view this entity as a hero. This final score for each entity is visualised in a plot (as in the image below). It indicates heroes by green bars and antiheroes by red bars. Entities regarded as neutral (an equal percentage of users view them as heroes and antiheroes) are marked by (short) blue bars.

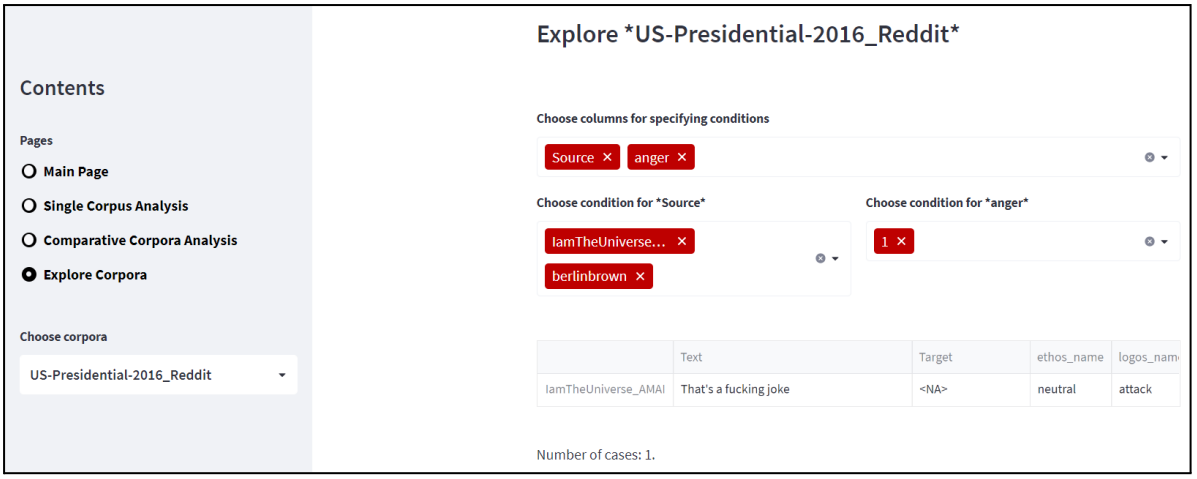


"Explore Corpora"

It allows you to explore a chosen dataset – see examples of cases in the corpora.

First, you can choose columns you want to specify conditions for (**Choose columns for specifying conditions**). Second, you set values for the columns you have chosen (**Choose condition for *[name of a column]***).

Based on those specified conditions (selected values for selected columns) the examples from the corpora are displayed.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Explore *US-Presidential-2016_Reddit*
Choose columns for specifying conditions
Source
anger
Choose condition for *Source*
IamTheUniverse...
berlinbrown
Choose condition for *anger*
1
Text
Target
ethos_name
logos_nam
IamTheUniverse_AMAI
That's a fucking joke
<NA>
neutral
attack
Number of cases: 1.